Current Group-I/II Chemistry and Charge-Neutrality Contract (v7.14.x)

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1. Status and authority

This document is the **accepted-current chemistry and charge-neutrality authority** for the v7.14.x grouped-site LTA cation generator. The historical `v7_13` filename records the release in which the Group-I/II extension was introduced; it does **not** limit the authority to v7.13.

For current releases:

- this document owns the supported cation set, formal charges, exact population semantics, and charge-neutrality invariant;
- `v7_14_group_i_ii_volume_reference_extension.md` owns the dry-volume reference model and empirical endpoint bindings;
- module specifications own their public software behavior and import these chemistry/volume semantics rather than redefining them;
- the architecture manual explains how the modules compose these authorities.

Earlier Li/Na/K-only statements are historical where they conflict with this contract.

2. Background and terminology

The supported framework is a conventional LTA aluminosilicate cell containing 24 framework Al, 24 framework Si, and 96 framework O atoms. Replacing a framework Si(IV) site by Al(III) introduces one net negative framework charge. The generator therefore balances the 24 framework-Al charge units with extra-framework cations.

For cation species i:

- N_i is its **exact nonnegative integer** population;
- z_i is its formal positive charge;
- $z_i N_i$ is its number of positive **charge equivalents**.

Charge equivalents are distinct from the number of occupied physical cation sites. A divalent cation occupies one physical site but supplies two charge equivalents.

3. Supported framework-compensating cations

The supported cation set is:

- Group I, formal charge +1: `H, Li, Na, K, Rb, Cs, Fr`;
- Group II, formal charge +2: `Be, Mg, Ca, Sr, Ba, Ra`.

The public constants are:

```
GROUP_I_CATIONS
GROUP_II_CATIONS
SUPPORTED_CATIONS
CATION_CHARGES
```

`ALLOWED_ALKALI` remains a backward-compatible alias for `SUPPORTED_CATIONS`; its historical name no longer limits the accepted species to alkali metals.

4. Current production framework scope

Production generation is intentionally restricted to:

```
Al24Si24O96
Si:Al = 1:1
framework negative charge = 24
```

Non-24-Al cells, other Si:Al ratios, arbitrary framework normalizations, and a generalized arbitrary-N_{Al} volume model are outside the v7.14.x production domain. Small reduced cells may exist as private algorithmic test fixtures, but they are not supported scientific generator inputs.

5. Exact population and charge-neutrality invariant

A production composition is admissible only when

$$\sum_i z_i N_i = 24.$$

Equivalent examples include:

```
24 Na+          -> 24 charge equivalents
12 Ca2+         -> 24 charge equivalents
8 Na+ + 8 Ca2+ -> 24 charge equivalents from only 16 occupied cation sites
```

The rule is therefore **not** "24 total cations." Group-II species always contribute +2.

Cation populations are discrete software inputs. Public/direct numerical interfaces must not silently convert a different numerical composition into an integer composition:

- Python and NumPy integer values are admissible when nonnegative;
- negative integers are rejected;
- floating values, including integer-valued floats such as `24.0`, are rejected where exact counts are required;
- Boolean values are rejected;
- textual CLI count specifications are parsed as integer counts and reject noninteger syntax.

Zero is an admissible per-species count, but the assembled composition must still satisfy the exact neutrality and site-capacity constraints.

6. Site-capacity and ratio semantics

Physical-site capacity is independent of formal charge. Each occupied grouped site holds one cation regardless of valence.

For every ring family:

- explicit counts are exact numbers of cations;
- ratio inputs are occupancy fractions of that ring-family physical-site capacity;
- ratios are converted to integer occupancies using the module's documented rounding behavior;
- exact charge neutrality is checked after the integer population is resolved.

7. Random, batch, maximum-symmetry, and smoothing modes

Every member of `SUPPORTED_CATIONS` is a valid occupation label in:

- random grouped-site population;
- symmetry-unique random batching;
- maximum-symmetry occupation optimization;
- exact-coordinate realization;
- symmetry-coupled smoothing and species exchange;
- density analysis and multi-format serialization.

The maximum-symmetry state alphabet is `empty` followed by `SUPPORTED_CATIONS` in the deterministic package order.

8. Serialization semantics

Serialized output groups atoms as `Si, Al, O`, followed by `SUPPORTED_CATIONS` in deterministic order.

LAMMPS data output uses `atom_style atomic`. It contains no per-atom force-field charge column. The formal charge table in this contract is used for chemical composition validation; it does not prescribe force-field partial charges.

9. Dry-volume authority

The chemistry contract deliberately does not define species-specific cell volumes. Current volume semantics, endpoint classifications, placeholder behavior, source transformations, and the fixed `/24` normalization are owned by:

`doc/specs/v7_14_group_i_ii_volume_reference_extension.md`.

10. Compatibility names

Legacy public names retained for compatibility include:

- `ALLOWED_ALKALI` -> all supported cations;
- `PopulationResult.alkali_counts` -> count map for all supported cations;
- `PopulationResult.required_cations` -> historical field containing the required positive charge;
- `ResolvedPopulationConstraints.total_alkali_count` -> alias of `total_cation_count`.

Preferred semantic names are `SUPPORTED_CATIONS`, `PopulationResult.cation_counts`, `PopulationResult.required_positi` and `ResolvedPopulationConstraints.total_cation_count`.

11. Historical note

v7.13 introduced the Group-I/II cation extension. That release temporarily used an input-template-volume fallback for species without the legacy Li/Na/K endpoints. v7.14 replaced that volume behavior with the current evidence-bound charge-equivalent model. This history is retained only to explain compatibility; it is not part of the present volume contract.